Estimated Annual Agricultural Pesticide Use
Klamath County, Oregon 2014-2018

24 July 2026

Summary

Klamath County ranks **#5 of Oregon's 36 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of OR's five highest pesticide-use counties. Klamath County, in the Klamath Basin, grows irrigated potatoes, alfalfa hay, and grain in a region long associated with disputes over irrigation water allocation.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Klamath County, Oregon** during 2018, as reported by USGS Estimated Annual Agricultural Pesticide Use data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by Thelin and Stone (2013) and Baker and Stone (2015):

> "EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD." *-Pesticide National Synthesis Project webpage*

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. "detailed interpretation of where and how much use occurs within a county is not appropriate").
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2) *(continued)*

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Klamath County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	DICHLOROPROPENE	OC	180927	145329	205443
2	METAM	DTC	152375	147779	133802
3	METAM POTASSIUM	DTC	66596	74946	80942
4	CHLOROPICRIN	OC	36821	16232	10755
5	SULFURIC ACID	INO	34849	78811	81322
6	GLYPHOSATE	OTH	27896	16538	23813
7	ETHOPROPHOS	OP	17915	9806	5884
8	TRIFLURALIN	OTH	12547	10358	17672
9	MANCOZEB	DTC	11155	6282	4932
10	EPTC	TC	4412	5534	7187
11	2,4-D	CPX	4004	4443	6200
12	DIURON	OTH	3334	2550	3381
13	PENDIMETHALIN	OTH	2175	1565	1816
14	OXAMYL	CB	2168	1727	1468
15	METRIBUZIN	TRZ	1972	1679	1988
16	BACILLUS AMYLOLIQUEFACIEN	MIC	1898	22358	22358
17	METOLACHLOR & METOLACHLOR-S	OTH	1795	1417	1725
18	METOLACHLOR-S	OTH	1786	1370	1660
19	BROMOXYNIL	OTH	1731	1631	1643
20	DIQUAT	DIP	1444	1103	1013
21	CHLORPYRIFOS	OP	1396	982	1388
22	MCPA	CPX	1198	2838	2391
23	PYMETROZINE	OTH	992	545	418
24	COPPER HYDROXIDE	INO	943	389	296
25	COPPER OXYCHLORIDE	INO	939	357	239
26	FLUOPYRAM	OTH	871	341	266
27	TERBACIL	OTH	820	820	625
28	SULFUR	INO	783	517	820
29	PYRIMETHANIL	OTH	774	406	430
30	PARAQUAT	DIP	694	897	1040
31	DICAMBA	OTH	692	633	794
32	AZOXYSTROBIN	OTH	585	486	537
33	MEFENOXAM	OTH	579	474	444
34	CLOPYRALID	OTH	562	350	434
35	FLONICAMID	OTH	505	374	326
36	FLUFENACET	OTH	377	284	304
37	FLUROXYPYR	OTH	366	600	608
38	METHOMYL	CB	350	310	506
39	DIMETHOATE	OP	342	501	667
40	PROPICONAZOLE	OTH	327	314	295
41	ATRAZINE	TRZ	272	447	316
42	FLUMIOXAZIN	OTH	271	118	97
43	PINOXADEN	OTH	256	152	164
44	TOLFENPYRAD	OTH	232	232	232
45	PYRACLOSTROBIN	OTH	209	267	283
46	PYRASULFOTOLE	OTH	202	151	120

Table 2: Top 80 EPest-high Estimates, Klamath County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) *(continued)*

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	IMAZETHAPYR	OTH	197	156	139
48	BIFENTHRIN	PYR	194	128	101
49	IMIDACLOPRID	OTH	191	319	357
50	HEXAZINONE	TRZ	168	953	860
51	CLETHODIM	OTH	162	270	334
52	PETROLEUM OIL	OTH	154	313	294
53	FLUAZINAM	OTH	148	155	293
54	CYHALOTHRIN-LAMBDA	PYR	133	130	145
55	BURKHOLDERIA SPP	MIC	130	2419	2419
56	ALPHA CYPERMETHRIN	PYR	130	71	71
57	ESFENVALERATE	PYR	130	105	80
58	PENTHIOPYRAD	OTH	120	127	137
59	IMAZAMOX	OTH	116	133	119
60	FLUXAPYROXAD	OTH	109	126	83
61	ZOXAMIDE	OTH	109	112	224
62	DIMETHENAMID & DIMETHENAMID-P	OTH	108	854	848
63	DIMETHENAMID-P	OTH	108	765	796
64	FLUPYRADIFURONE	OTH	108	41	117
65	THIFENSULFURON	OTH	103	103	94
66	TRIBENURON METHYL	OTH	92	87	81
67	2,4-DB	CPX	85	805	1168
68	CYFLUTHRIN	PYR	80	103	130
69	FLUTOLANIL	OTH	77	290	454
70	BOSCALID	OTH	75	110	184
71	GLUFOSINATE	OTH	69	227	171
72	MALEIC HYDRAZIDE	OTH	67	137	161
73	CYMOXANIL	OTH	60	51	84
74	DIFENOCONAZOLE	OTH	58	149	122
75	FAMOXADONE	OTH	57	50	83
76	SPIROTETRAMAT	OTH	54	107	115
77	FIPRONIL	OTH	45	44	70
78	TEBUCONAZOLE	OTH	42	50	42
79	PROTHIOCONAZOLE	OTH	41	23	34
80	BENTAZONE	OTH	38	49	55

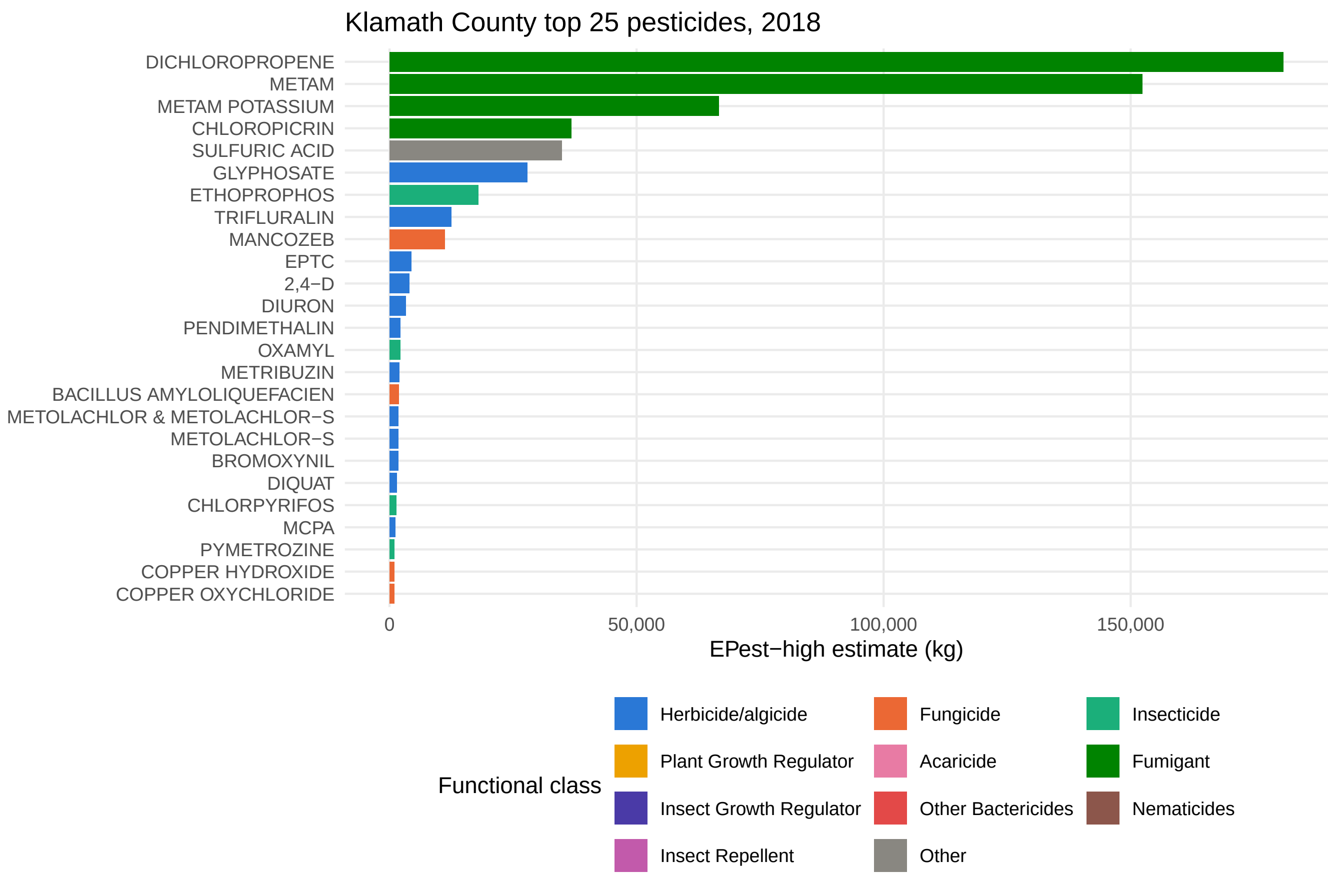

Figure 1: Top 25 compounds in Klamath County by EPest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Klamath County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	DICHLOROPROPENE	180927	180927
2	METAM	1078	152375
3	METAM POTASSIUM	0	66596
4	CHLOROPICRIN	1534	36821
5	SULFURIC ACID	0	34849
6	GLYPHOSATE	26309	27896
7	ETHOPROPHOS	1	17915
8	TRIFLURALIN	16	12547
9	MANCOZEB	11155	11155
10	EPTC	2	4412
11	2,4-D	3968	4004
12	DIURON	404	3334
13	PENDIMETHALIN	2175	2175
14	OXAMYL	1	2168
15	METRIBUZIN	989	1972
16	BACILLUS AMYLOLIQUEFACIEN	0	1898
17	METOLACHLOR & METOLACHLOR-S	98	1795
18	METOLACHLOR-S	88	1786
19	BROMOXYNIL	316	1731
20	DIQUAT	1444	1444
21	CHLORPYRIFOS	1391	1396
22	MCPA	328	1198
23	PYMETROZINE	992	992
24	COPPER HYDROXIDE	27	943
25	COPPER OXYCHLORIDE	22	939
26	FLUOPYRAM	871	871
27	TERBACIL	820	820
28	SULFUR	783	783
29	PYRIMETHANIL	774	774
30	PARAQUAT	327	694
31	DICAMBA	31	692
32	AZOXYSTROBIN	360	585
33	MEFENOXAM	578	579
34	CLOPYRALID	7	562
35	FLONICAMID	505	505
36	FLUFENACET	0	377
37	FLUROXYPYR	4	366
38	METHOMYL	58	350
39	DIMETHOATE	288	342
40	PROPICONAZOLE	68	327
41	ATRAZINE	272	272
42	FLUMIOXAZIN	6	271
43	PINOXADEN	0	256
44	TOLFENPYRAD	0	232
45	PYRACLOSTROBIN	8	209
46	PYRASULFOTOLE	0	202
47	IMAZETHAPYR	197	197
48	BIFENTHRIN	10	194
49	IMIDACLOPRID	8	191
50	HEXAZINONE	168	168

Table 4: Top 50 Klamath County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	97347	147779
2	DICHLOROPROPENE	117780	145329
3	SULFURIC ACID	0	78811
4	METAM POTASSIUM	0	74946
5	BACILLUS AMYLOLIQUEFACIEN	0	22358
6	GLYPHOSATE	12299	16538
7	CHLOROPICRIN	537	16232
8	TRIFLURALIN	22	10358
9	ETHOPROPHOS	907	9806
10	MANCOZEB	6282	6282
11	EPTC	562	5534
12	2,4-D	4150	4443
13	QUINTOZENE	0	2898
14	MCPA	700	2838
15	DIURON	479	2550
16	BURKHOLDERIA SPP	0	2419
17	PHOSPHORIC ACID	0	1875
18	OXAMYL	924	1727
19	METRIBUZIN	1330	1679
20	BROMOXYNIL	697	1631
21	PENDIMETHALIN	1276	1565
22	CHLOROTHALONIL	1445	1518
23	METOLACHLOR & METOLACHLOR-S	229	1417
24	METOLACHLOR-S	181	1370
25	DIQUAT	1103	1103
26	CHLORPYRIFOS	939	982
27	HEXAZINONE	304	953
28	NALED	0	922
29	PARAQUAT	414	897
30	ETHALFLURALIN	38	873
31	DIMETHENAMID & DIMETHENAMID-P	157	854
32	TERBACIL	820	820
33	2,4-DB	0	805
34	MALATHION	38	767
35	DIMETHENAMID-P	68	765
36	DICAMBA	213	633
37	FLUROXYPYR	188	600
38	PYMETROZINE	545	545
39	SULFUR	517	517
40	DIMETHOATE	103	501
41	AZOXYSTROBIN	203	486
42	MEFENOXAM	469	474
43	ATRAZINE	436	447
44	PYRIMETHANIL	404	406
45	COPPER HYDROXIDE	40	389
46	FLONICAMID	354	374
47	COPPER OXYCHLORIDE	8	357
48	CLOPYRALID	8	350
49	FLUOPYRAM	340	341
50	IMIDACLOPRID	258	319

Table 5: Top 50 Klamath County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	DICHLOROPROPENE	188913	205443
2	METAM	82717	133802
3	SULFURIC ACID	0	81322
4	METAM POTASSIUM	618	80942
5	GLYPHOSATE	17435	23813
6	BACILLUS AMYLOLIQUEFACIEN	0	22358
7	TRIFLURALIN	16	17672
8	CHLOROPICRIN	1338	10755
9	EPTC	1752	7187
10	2,4-D	5976	6200
11	ETHOPROPHOS	544	5884
12	MANCOZEB	4932	4932
13	PHOSPHORIC ACID	6	4407
14	DIURON	401	3381
15	QUINTOZENE	0	3006
16	METHYL BROMIDE	2695	2695
17	BURKHOLDERIA SPP	0	2419
18	MCPA	712	2391
19	METRIBUZIN	1542	1988
20	PENDIMETHALIN	1469	1816
21	MALATHION	46	1764
22	METOLACHLOR & METOLACHLOR-S	1001	1725
23	METOLACHLOR-S	943	1660
24	BROMOXYNIL	712	1643
25	CHLOROTHALONIL	1384	1470
26	OXAMYL	982	1468
27	CHLORPYRIFOS	920	1388
28	2,4-DB	0	1168
29	METIRAM	0	1151
30	PARAQUAT	559	1040
31	DIQUAT	1013	1013
32	NALED	0	922
33	HEXAZINONE	274	860
34	DIMETHENAMID & DIMETHENAMID-P	379	848
35	SULFUR	820	820
36	DIMETHENAMID-P	326	796
37	DICAMBA	364	794
38	DIMETHOATE	143	667
39	TERBACIL	410	625
40	FLUROXYPYR	318	608
41	ETHALFLURALIN	59	560
42	AZOXYSTROBIN	201	537
43	METHOMYL	225	506
44	FLUTOLANIL	301	454
45	MEFENOXAM	405	444
46	CLOPYRALID	5	434
47	PYRIMETHANIL	426	430
48	PYMETROZINE	393	418
49	IMIDACLOPRID	311	357
50	CLETHODIM	139	334